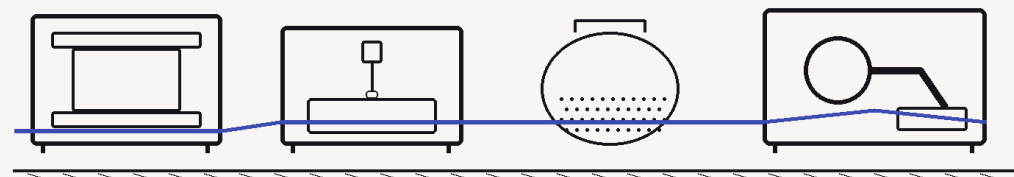


Five material classes could unlock **5–12 GtCO₂/year** — if the predictions sent to laboratories are corrected and verified first.

Cobalt-free cathodes. Halide solid electrolytes. Direct-air-capture sorbents. Ammonia catalysts. Lead-free perovskites. The opportunity is climate-scale; the bottleneck is whether simulation results are reliable enough to spend scarce laboratory time on.

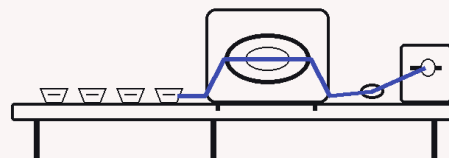


380,000 computationally stable structures; 736 independently synthesized by late 2023: a reported 0.2% validation rate.

380,000 736

REPORTED 0.2% VALIDATION RATE

Generation now outruns validation. At climate scale, more candidates do not help if errors survive generation, prediction, synthesis, and validation.



Validation-rate indicator; not a universal synthesizability probability.

Foundation MLIPs were measured across **21 materials**,
up to 9 properties, and **228 published reference values**.

21

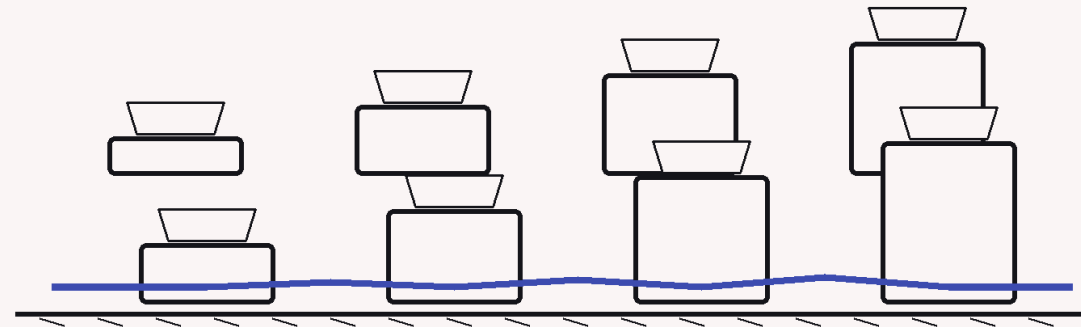
MATERIALS

up to 9

PROPERTIES

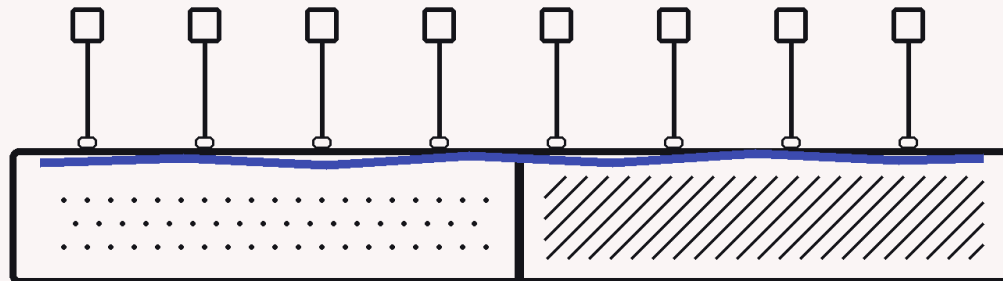
228

REFERENCE VALUES



In the measured benchmark, rankings often survive even when hard-property magnitudes miss by tens of percent. Bounded to the measured benchmark domain.

A measured physical error field runs beside the live MLIP as an additive energy term with analytic forces. **No fine-tuning. No retraining.**



EXISTING MLIP

↓ MEASURED ROUTE

MEASURED ERROR FIELD

↓ ADDITIVE TERM

CORRECTED ENERGY + ANALYTIC FORCES

NO FINE-TUNING / NO RETRAINING

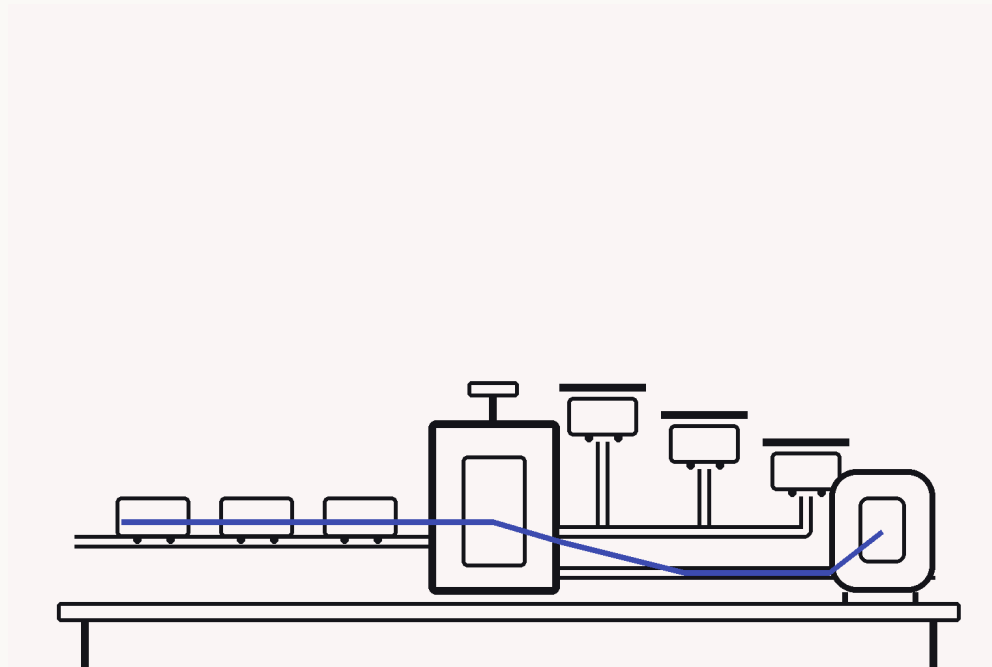
Every prediction takes one of three routes: **measured correction**, **bounded claim**, or **abstention** when evidence is insufficient.

MEASURED CORRECTION

BOUNDED CLAIM

ABSTAIN

The proof layer licenses corrections only where evidence supports them. The honesty proof is preserved in public: the first correction made alloy predictions worse; the resulting Lean law now blocks wrong-direction correction, and unsupported cases abstain.



The first correction made predictions worse; the Lean law blocks wrong-direction correction.

129/129 anchors evaluated. Zero evaluation failures. 61.0 wall-hours. \$14.65 cloud-equivalent.

129/129

ANCHORS EVALUATED

zero

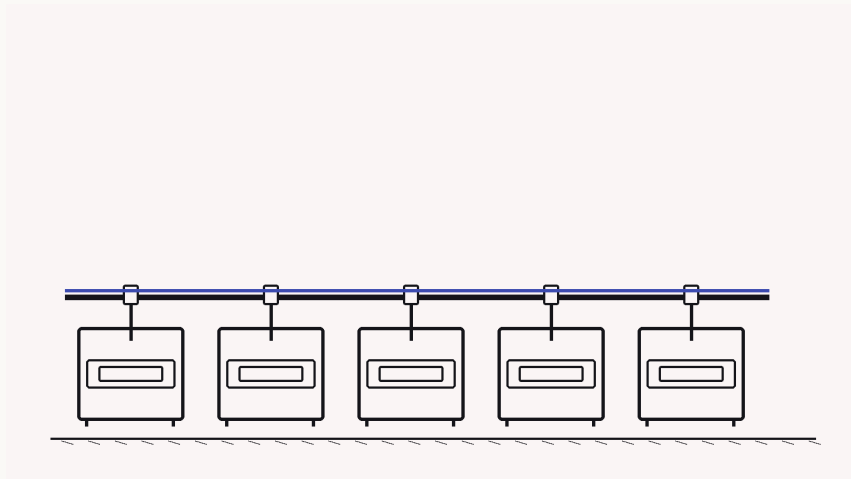
EVALUATION FAILURES

61.0

WALL-HOURS

\$14.65

CLOUD-EQUIVALENT



The evidence program spans **42 published Li-ion electrolyte chemistries**: a **30-path frozen test panel** plus **12 disjoint training chemistries**, evaluated with guidance from **4 foundation MLIPs**.

Two guidance models found the true extrema on every guided path; two recorded a 6.8 meV undercoverage deficit. FOR EDITOR REVIEW; not peer-reviewed.

Across the locked panel's **29 analyzable paths**, sharing anchors across 4 models reduced the recomputed requirement from **558 to 154 DFT evaluations**: 72.4% fewer, or 3.62×.

RECOMPUTED / 29 ANALYZABLE PATHS

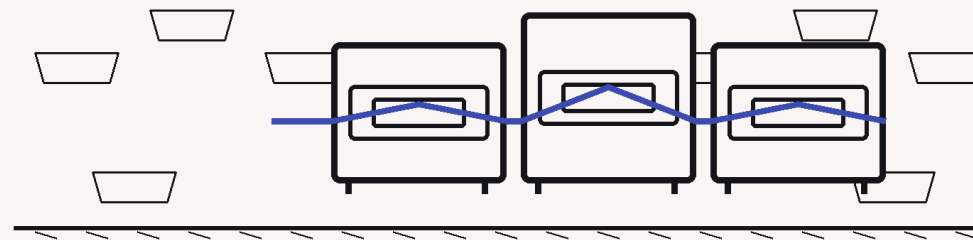
558

154

72.4% FEWER / 3.62×

The naive bill scales from **139.5 to 558** as guidance grows from one model to four; the shared union grows from **139.5 to 154**, about **10%**. No new DFT was run to produce this recomputation.

RECOMPUTED 558→154 ≠ EXECUTED CAMPAIGN RECORD



Physical-law theorems are non-rival: partners can contribute verified laws without disclosing proprietary chemistry, and every participant's gates sharpen.

The evidence chain ties quantitative claims to provenance-hashed data and machine-checked Lean theorems. Each failed correction and impossibility result sharpens the boundary rather than disappearing from the record.

100/100 unique still slots

33 certified baseline stills + 67 independently certified Wave-4 replacements

Closure gate t_c2a1f8e3 · five certified films excluded from the denominator



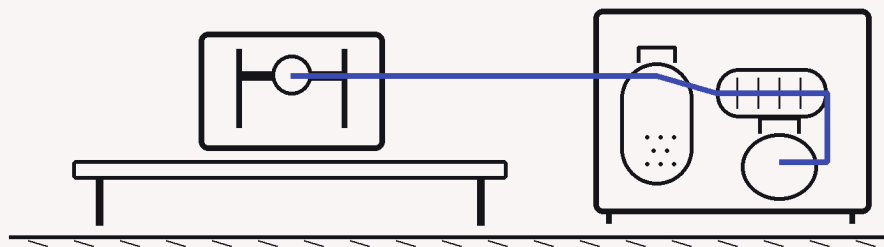
Lupine sits between model generation and laboratory validation: model builders get external verification; laboratories get shortlists with characterized failure modes.

MODEL GENERATION

LUPINE TRUST LAYER

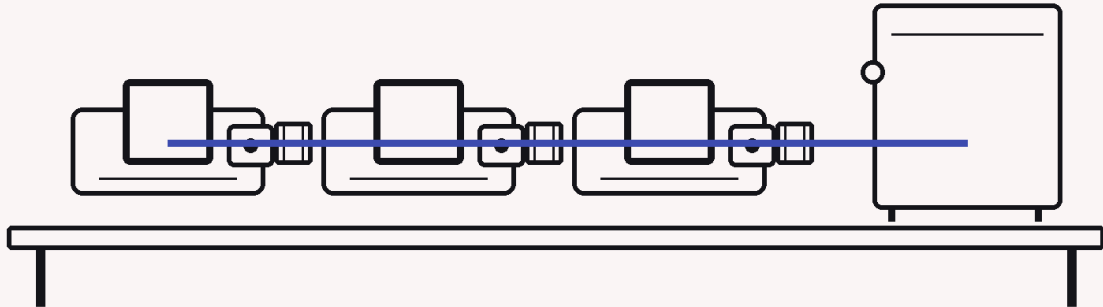
LABORATORY VALIDATION

The first market wedge is teams already running MLIP simulation in climate-critical materials. The product complements generators, databases, simulators, autonomous labs, and experimental partners rather than replacing them.



A frozen public panel, four-model guidance, a completed 129-anchor execution record, and a reproducible 558 → 154 savings analysis are inspectable now.

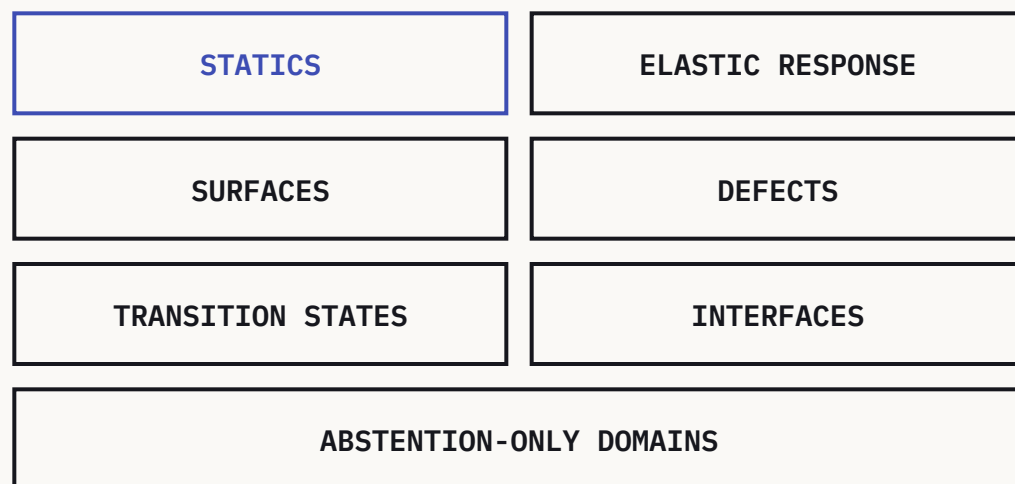
frozen public panel	INSPECTABLE NOW
four-model guidance	INSPECTABLE NOW
completed 129-anchor execution record	INSPECTABLE NOW
reproducible 558 → 154 savings analysis	INSPECTABLE NOW



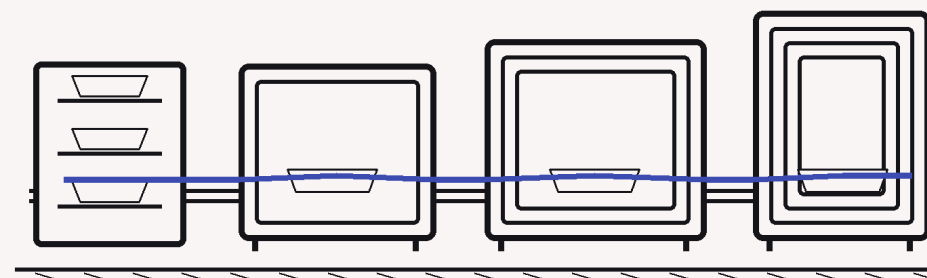
Completed evidence infrastructure – not commercial revenue or peer-reviewed validation. Z1 execution source: FOR EDITOR REVIEW – not for citation.

Lupine Science, “The Z1 Barrier Panel,” lines 40–66; “The Union Verdict,” lines 17–35 and 75–82 (FOR EDITOR REVIEW – not for citation); “The Savings Stack,” lines 26–56.

Advance by order of effort: **statics** → elastic response → surfaces → defects → transition states → interfaces → abstention-only domains.

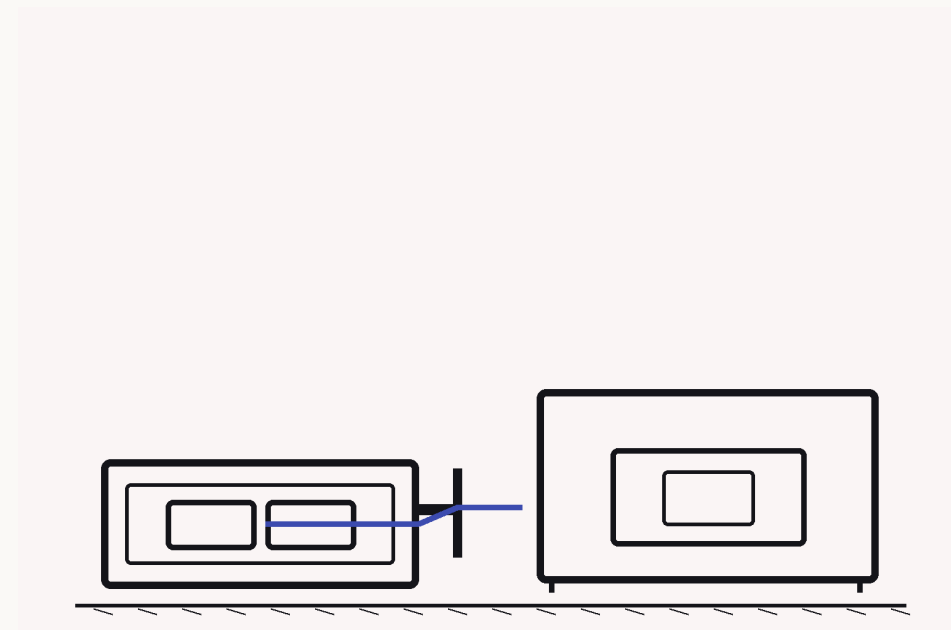


Instrument the easiest proven runtime lane; extend structure-family jurisdiction only behind explicit gates; use sparse DFT anchors where correction laws cannot act; abstain where no supported measurement path exists.



One 30-path panel. One chemistry family. Not peer-reviewed.

- one DFT engine and functional in the executed Z1 record
- short paths make sparsity easier
- seven large paths remain deferred
- two models recorded guidance misses
- no GPAW↔VASP equivalence claim



Build correction-as-acceleration into the simulation loop used by every materials lab.

[OWNER DECISION] Round size
[OWNER DECISION] Instrument / terms
[OWNER DECISION] Runway and allocation

USE-OF-FUNDS FRAME

Runtime instrumentation in the proven lane.

Gated expansion across harder structure and property families.

Partner-facing evidence delivery from simulation to synthesis.

Models guide. Evidence measures.
Proof keeps the score.

